

Vitchakorn Poonyakanok, MD

Physician & Clinical Data Analyst

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Professional Profile

Physician-technologist working at the intersection of frontline clinical practice and data science — a bridge between the two worlds. I turn clinical and public-health problems into working software and analytics: a deployed altitude-risk decision-support app ([altit.app](#)), field outbreak investigations, predictive clinical models, and edge-AI health devices. I bring clinical credibility to data and engineering teams, and engineering rigour (ML, data governance, interoperability, cybersecurity) to health systems.

Education

- Aug 2026 – 2028 **M.S. in Health Care Analytics & Information Technology (MSHCA)**, *Heinz College, Carnegie Mellon University, Pittsburgh, PA, USA*, Royal Thai Government (DDC) scholarship
Focus: health system analytics, data governance, and information systems.
- Aug 2025 – Jun 2026 **M.S. in Digital & AI Technologies in Health Systems (DigiHealth)**, *Faculty of Medicine & Faculty of Engineering, Chulalongkorn University, Bangkok, Thailand*
Focus: digital health architecture, clinical AI applications, interoperability, and medical databases.
Independent study: *Development and Evaluation of an AI-Assisted Altitude Extraction System from Travel Itineraries for Travel Medicine* (advisor: Assoc. Prof. Watcharapong Piyaphanee, M.D.).
- Jun 2024 – Jul 2026 **Residency Training in Preventive Medicine (Travel Medicine)**, *Institute of Preventive Medicine, DDC, Ministry of Public Health, Nonthaburi, Thailand*
Focus: preventive medicine, travel risk assessment, vaccinology, and health security.
On planned academic leave from Aug 2026 to undertake the MSHCA at Carnegie Mellon; residency to resume on return.
- Apr 2013 – Jul 2019 **Doctor of Medicine (MD)**, *Faculty of Medicine Ramathibodi Hospital, Mahidol University, Bangkok, Thailand*
Clinical rotations across Internal Medicine, Surgery, Pediatrics, OB-GYN, and Emergency Medicine.
International clinical elective via the IFMSA SCOPE exchange — Hospital Clínico Universidad de Chile, Santiago (2017).
- 2010 – 2013 **Upper Secondary School (Science–Mathematics Program)**, *Triam Udom Suksa School, Bangkok, Thailand*

Work Experience

- Nov 2024 – Jul 2026 **General Practitioner — Preventive & Travel Medicine**, 🏢 *Office of Disease Prevention and Control 6 (ODPC 6), DDC, MOPH, Pattaya, Chon Buri, Thailand*
 - Deliver pre-travel risk assessment, vaccination, and prophylaxis counselling; apply self-built clinical tools (e.g. altitude-risk analysis) at the point of care.
 - Serve as GP and site physician for the WHO Enhanced Gonococcal Antimicrobial Surveillance Programme (EGASP), contributing standardized surveillance data.*On academic leave from Aug 2026 for the MSHCA at Carnegie Mellon.*
- Jun 2022 – May 2024 **General Practitioner**, 🏢 *Bamrasnaradura Infectious Diseases Institute, Nonthaburi, Thailand*
 - Led outpatient and Acute Respiratory Infection (ARI) clinic services through major COVID-19 and mpox outbreak responses.
 - Analyzed outpatient flow logs in Python to identify clinic bottlenecks, and presented the findings and scheduling recommendations to department leadership.
 - Delivered vaccinology counselling and designed clinical safety protocols for staff.

- Jul 2020 – **General Practitioner (Intern)**, 🏠 *Si Songkhram Hospital*, Nakhon Phanom, Thailand
- Apr 2022 ○ Managed general, emergency, and inpatient medical services at a community hospital.
- Sep 2019 – **General Practitioner (Intern)**, 🏠 *Nong Khai Hospital*, Nong Khai, Thailand
- Jul 2020 ○ Completed intern clinical rotations in emergency medicine, pediatrics, surgery, and OB-GYN.

Publications & Outbreak Investigations

- Peer-reviewed **Investigation of MMR Vaccine-Related Mumps Infections Cluster in Practical Nurse Students, Bangkok, Thailand** (2025). S. Lertsakulbunlue, D. Srithammavong, K. Tepsittha, H. Kanjanasombut, V. Poonyakanok, et al. *Journal of Tropical Medicine*. (co-author; field outbreak investigation)
- Surveillance evaluation **Evaluation of the Syphilis Surveillance System**, ODPC 6, Pattaya, Chon Buri, Thailand (Jan–Jun 2025). Assessed the national Digital Disease Surveillance (DDS) system’s attributes — sensitivity, predictive value positive, timeliness, and representativeness — and recommended improvements. (co-investigator; preventive-medicine residency study)
- Surveillance report **Preliminary Investigation and Rapid Response to a Methanol Intoxication Outbreak Linked to Illegal Herbal Liquor in Bangkok, August 2024**. C. Jitpeera, W. Rongdech, K. Raruenroeng, V. Poonyakanok, et al. *Weekly Epidemiological Surveillance Report (WESR)*, 2024. (co-author; rapid outbreak response)

Applied AI & Technical Training

- Apr – Jun 2026 **Super AI Engineer Season 6**, *Artificial Intelligence Association of Thailand (AIAT)*, Thailand, Advanced AI Level 2 — selected as one of the top 150 from 10,000+ applicants
 - **Coffee Chain Demand Forecasting Hackathon**: Private Kaggle Rank #1 / 2nd Runner-up overall (of 30 teams) — leakage-aware time-series pipeline with gradient boosting and robust cross-validation.
 - **Parasite Eggs Object Detection Challenge**: Ranked 16/166 — cascaded microscopy detector using SAHI, YOLOv8, and a DINOv2 classification head.
 - **Edge-AI Intelligent Transport System Hackathon**: Co-Honorable Mention — real-time CCTV edge analytics on Jetson Nano (YOLO/ONNX Runtime).
 - **WellSense AIoT Product Hackathon**: Academic Popular Vote — wearable running-posture monitor (Arduino Nano 33 BLE Sense, TinyML/Edge Impulse, BLE, React dashboard).
 - **Agentic Pipeline Hackathon (2026 Thai Election OCR)**: Ranked 102/571 — OCR pipeline for hand-written election reports.

Selected Projects

- Deployed tools
 - **Altitude Itinerary Analyzer** ([altit.app](#)) — full-stack clinical decision support web app that parses and analyzes multi-modal travel itineraries for high-altitude sickness risk assessment.
 - **Thai HospSearch** ([hoscode.vercel.app](#)) — interactive lookup mapped to the Thai healthcare administrative hierarchy (Region to Subdistrict) for administrative coding and records checking.
 - **Hajj Menstrual Delay Planner** ([hajj-period-planner.vercel.app](#)) — clinical utility for planning pharmaceutical menstrual suppression for female Hajj pilgrims.
- Data science
 - **Cirrhosis Survival Prediction** ([Kaggle](#)) — multi-class survival model using ensemble learning (XGBoost/LightGBM).
 - **Outpatient Waiting Time Analysis** ([Kaggle](#)) — ANOVA and Random Forest on hospital logs to pinpoint bottlenecks and forecast wait times.
 - **Sichuan-Yunnan Altitude & Transit Map** ([GitHub](#)) — QGIS geospatial visualization for Acute Mountain Sickness risk along the Sichuan-Yunnan highway.
- Dashboards
 - **Tableau: PM2.5 & Mental Health Dynamics** (with Interrupted Time Series), **Yellow Fever & Malaria Prevention**, and **Thai EPI Vaccine Dashboard**.

Certifications

Security & Cloud

- TPQI × Huawei — AI Literacy Power Users & Cloud Developer Level 5 — 2026.
- ISC2 Certified in Cybersecurity (CC) — Apr 2026–Mar 2029.
- Huawei Cloud HCCDA — Tech Essentials & AI — Apr 2026–Apr 2029.
- Cybersecurity Foundation Course, Thailand National Cyber Academy — 2024.
- Google Cybersecurity Professional Certificate — 2024.

Data & AI

- Super AI Engineer — AI Practitioner Certification (Level 1), AIAT — 2026; includes Foundation AI (Theory), From Data to Insight, and the FahMai RAG & Agentic Pipeline hackathons.
- Google AI Essentials V1 — 2024.
- Certified Professional Data Analyst, DataRockie Bootcamp 9 — 2024.
- Google Advanced Data Analytics Professional Certificate — 2023.
- Data Visualization with Tableau Specialization, UC Davis — 2023.
- Excel Level Up Update 2021, SkillLane — 2023.
- Google Data Analytics Professional Certificate — 2023.

Medical & Research

- CITI Program Biomedical Researchers — Jan 2025–Jan 2029.
- Introductory Course in Epidemiology & Biostatistics, DDC MOPH — 2024.
- ISTM Review Course, International Society of Travel Medicine — 2024.
- Diploma in Clinical Epidemiology, Chiang Mai University — Sep–Dec 2022.
- Diploma in Clinical Statistics, Chiang Mai University — Feb–May 2022.

Licensure & Registration

Medical License, Thai Medical Council — full registration since 2019.

Tour Leader License, Department of Tourism, Thailand — 2025.

Technical Skills

Programming & Statistics: Python, R, SQL, STATA

Machine Learning & Data Science: Scikit-Learn, gradient boosting & stacked ensembles (XGBoost, LightGBM, CatBoost), AutoGluon (AutoML), Chronos (time-series), PyTorch, YOLO, OpenCV, Pandas, NumPy, Jamovi

LLM & Generative AI: Claude, GPT, Gemini, GLM, Grok, and ThaiLLM APIs; RAG and agentic pipelines

Visualization & Geospatial: Tableau, Looker, Power BI, Matplotlib, Seaborn, QGIS, Adobe Photoshop, Inkscape

Software, Cloud & Hardware: Git, Firebase, Google Cloud Platform (GCP), Huawei Cloud, Arduino, TinyML, UML

Environments & HPC: Colab, Jupyter, Anaconda, Linux, LANTA HPC

Clinical Data & Interoperability: data interoperability, data governance, medical record systems

Languages & Interests

Languages Thai (Native); English — Proficient (TOEFL iBT 91/120, Duolingo English Test 120/160).

Medical Travel medicine, preventive vaccinology, emerging infectious diseases.

Technical Public health informatics, clinical machine learning, geospatial disease mapping, healthcare cybersecurity.

Personal Scuba diving (PADI Rescue Diver), photography, trekking, and travel.